INDIVIDUAL EXTENSION / UW IS 445

Gen Z spending: re-checking a course project

Individual extension of a UW IS 445 group project (2025)

1,700
financial records, ages 18-27

0.03
correlation: income vs consumption

50.6%
spend more than they earn

Both numbers match the original course presentation. Rebuilt in SQLite and checked independently in Python.

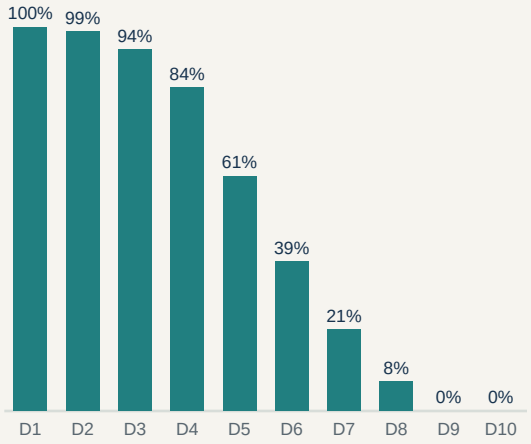
What the re-check found

1. **The financial data looks randomly generated**
   All 14 numeric columns are spread evenly between round limits (income: $1,501 to $7,994), and no two columns correlate above 0.06. The near-zero correlation reflects how the file was built, not how Gen Z spends.

2. **Spending above income is mostly an income effect**
   Average consumption is about $4,800 at every income level, so 88% of below-median earners spend more than they earn, versus 14% of everyone else.

3. **Age groups shop the same way (synthetic data)**
   In a separate synthetic shopping dataset, 0 of 4 comparisons (payment, shipping, frequency, category) differ between ages 18-27 and 28-70 after Holm correction.

Spending more than income, by income decile



Lowest to highest income

What I would do with real customer data

- **Segment real customers on what they actually buy (category mix, repeat purchases, response to past promotions), not on income or age alone.**
  Income and consumption are unrelated in this file (r = 0.03); average consumption stays between $4,660 and $4,866 in every income decile.

- **Start with one shared offer, then A/B test single changes (price, delivery fee, promotion timing) on small groups before rolling them out.**
  In the synthetic shopping data, 0 of 4 comparisons (payment, shipping, frequency, category) differ between ages 18-27 and 28-70 after correction (Cramer's V at most 0.04).

- **Check where a dataset comes from and how it is distributed before building recommendations on it.**
  A quick distribution check showed every numeric column spread evenly between round limits, with no pair of columns correlated above 0.06.

Limits: the two datasets are separate and never joined; amounts are USD with no stated period; results describe these files, not real Gen Z consumers.

Sources: Financial v1 (2025-01-31, CC0) | Shopping v2 (synthetic)

Financial dataset | Synthetic shopping dataset

Course context: UW IS 445 group project, March-June 2025. My role: data analysis and visualization with SQL, Excel and Python. I rebuilt this extension on my own in October 2026, after the original course files were lost.